

The mysterious universe: Summary

Student: Class:

Use the words listed below to complete the following sentences.

absolute	protostar	electrons	spiral	observer
constellations	space	giants	billion	radio
fusion	time	nuclei	expansion	temperature
nebulae	afterglow	pulsars	helium	

- Groups of stars form patterns that are called
- Our solar system is part of a large galaxy called the Milky Way.
- Clouds of dust and gas in interstellar space are called
- The apparent movement of stars at different distances is caused by the movement of the This effect is called parallax.
- According to the big bang theory, the universe is about 13.7 years old.
- The first stage of the big bang involved the formation of and space. The early universe was very small and very hot. Space then underwent rapid and it cooled as it expanded.
- Quarks and were the first matter particles to form. Subsequent cooling led to proton and neutron formation, which then combined to form
- Red shifting of lines in star spectra is consistent with the stretching of as the universe expands.
- The big bang theory can explain the large amount of relative to hydrogen in the universe.
- Microwave background radiation has been detected by instruments aboard satellites. This microwave radiation is the cooled of the big bang.
- Stars are formed when gravity causes gas and dust to come together and heat up. Eventually a is formed and then finally a star of a given mass.
- Stars like our sun evolve into red before they explode.
- Red supergiants evolve into black holes or
- The magnitude of a star is a measure of its real brightness when all stars are compared at the same distance.
- The Hertzsprung–Russell diagram plots the surface of the star versus its absolute magnitude. This diagram helps us understand the evolutionary life cycle of a star.
- Stars generate their energy by nuclear in which hydrogen nuclei join together to form heavier nuclei.
- The universe can be studied using optical (visible light) telescopes as well as telescopes.